

$$*) \int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)} dx = ?$$

Pinar Albayrak
Final Uygulama

$$\frac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3}$$

$$A = \frac{x^2+4x+1}{(x+1)(x+3)} \Big|_{x=1} = \frac{3}{4}$$

$$B = \frac{x^2+4x+1}{(x-1)(x+3)} \Big|_{x=-1} = \frac{1}{2}$$

$$C = \frac{x^2+4x+1}{(x-1)(x+1)} \Big|_{x=-3} = -\frac{1}{4}$$

$$I = \frac{3}{4} \int \frac{dx}{x-1} + \frac{1}{2} \int \frac{dx}{x+1} - \frac{1}{4} \int \frac{dx}{x+3}$$

$$= \frac{3}{4} \ln|x-1| + \frac{1}{2} \ln|x+1| - \frac{1}{4} \ln|x+3| + C$$

$$*) \int \frac{-2x+4}{(x^2+1)(x-1)^2} dx = ?$$

$$\frac{4-2x}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$$

$$4-2x = (A+C)x^3 + (-2A+B-C+D)x^2 + (A-2B+C)x + (B-C+D)$$

$$A+C=0 \quad (1)$$

$$-2A+B-C+D=0 \quad (2)$$

$$A-2B+C=-2 \quad (3)$$

$$B-C+D=4 \quad (4)$$

$$(4) + (2) \Rightarrow \boxed{A=2} \rightarrow C=-A \Rightarrow \boxed{C=-2}$$

$$(3) \rightarrow B = \frac{A+C+2}{2} = \underline{\underline{1}}$$

$$(4) \rightarrow D = 4 - B + C = \underline{\underline{1}}$$

$$\int \frac{4-2x}{(x^2+1)(x-1)^2} dx = \int \frac{2x+1}{x^2+1} dx - 2 \int \frac{dx}{x-1} + \int \frac{dx}{(x-1)^2}$$

$$\underbrace{\int \frac{2x}{x^2+1} dx}_{\ln|x^2+1|} + \underbrace{\int \frac{1}{x^2+1} dx}_{\text{Arctan } x} - 2 \ln|x-1| - \frac{1}{x-1} + C$$

$$= \ln|x^2+1| + \text{Arctan } x - 2 \ln|x-1| - \frac{1}{x-1} + C$$

$$(*) I = \int e^{2x} \cos 3x dx = ?$$

$$\begin{aligned} \cos 3x = u &\rightarrow -3 \sin 3x dx = du \\ e^{2x} dx = dv &\rightarrow v = \frac{e^{2x}}{2} \end{aligned} \quad \left\{ \begin{aligned} \sin 3x = u &\rightarrow 3 \cos 3x dx = du \\ e^{2x} dx = dv &\rightarrow v = \frac{e^{2x}}{2} \end{aligned} \right.$$

$$I = \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x dx = \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \left(\frac{e^{2x} \sin 3x}{2} - \frac{3}{2} \int e^{2x} \cos 3x dx \right)$$

$\underbrace{\hspace{10em}}_I$

$$I = \frac{1}{2} e^{2x} \cos 3x + \frac{3}{4} e^{2x} \sin 3x - \frac{9}{4} I$$

$$\frac{13}{4} I = \frac{1}{2} e^{2x} \cos 3x + \frac{3}{4} e^{2x} \sin 3x \Rightarrow I = \frac{2}{13} e^{2x} \cos 3x + \frac{3}{13} e^{2x} \sin 3x + C$$

$$(*) \int \frac{2x+1}{x^3-7x+6} dx$$

$$\left. \begin{array}{c|ccc} 1 & 1 & 0 & -7 & 6 \\ & & 1 & 1 & -6 \\ \hline & 1 & 1 & -6 & 0 \\ \hline & x^2+x-6=0 & & & \\ & (x-2)(x+3) & & & \end{array} \right\}$$

$$x^3-7x+6 = (x-1)(x-2)(x+3)$$

$$\frac{2x+1}{x^3-7x+6} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$$

$$A = \frac{2x+1}{(x-2)(x+3)} \Big|_{x=1} = -\frac{3}{4}$$

$$B = \frac{2x+1}{(x-1)(x+3)} \Big|_{x=2} = 1$$

$$C = \frac{2x+1}{(x-1)(x-2)} \Big|_{x=-3} = -\frac{1}{4}$$

$$\int \frac{2x+1}{x^3-7x+6} dx = -\frac{3}{4} \int \frac{dx}{x-1} + \int \frac{dx}{x-2} - \frac{1}{4} \int \frac{dx}{x+3}$$

$$= -\frac{3}{4} \ln|x-1| + \ln|x-2| - \frac{1}{4} \ln|x+3| + C$$

21.12.2018

Sevgili Matematik 1 Öğrencileri,

Dönemin bu son notunu hazırlarken son kez yazmak istedim sizlere. Öncelikle, yayınladığım bu notlar sebebiyle teşekkür maili atan, odama ziyarete gelen, öğrencim olan arkadaşıyla selam yollayan, içinden iyi dilekler/dualar geçiren tüm öğrencilere teşekkür ederim ☺ Yazdığınız/söylediğiniz kadar fayda sağladıysa notlarım sizlere, yükünüzü bir nebze olsun azaltabildiysem ne mutlu bana ☺

Biliyorum sınavdan sınava koşturmaktan çok yoruldunuz/bunaldınız; ancak son dönemece girdiniz, ufukta tatil göründü ☺ Son bir gayretle final sınavlarınıza güzelce çalışın ki bütünlemelere kalmadan ailenizin yanına koşun...

Hepinize yaklaşan finallerinizde başarılar dilerim, umarım emeğinizin karşılığını alacağınız sorular ve tabii notlarla karşılaşacaksınız...

Gelecek sene bu minik notları okumak zorunda kalmamanız dileğiyle ... ☺

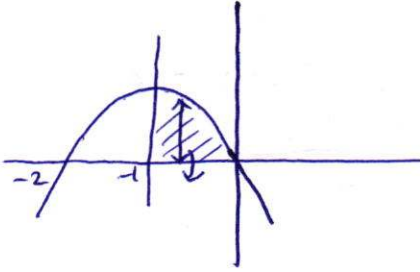
Şimdiden iyi tatiller,

Sevgiler,

Pınar Albayrak

4.a. $y = -x^2 - 2x$, $x \geq -1$ ve $y = 0$ ile sınırlı olan bölgenin

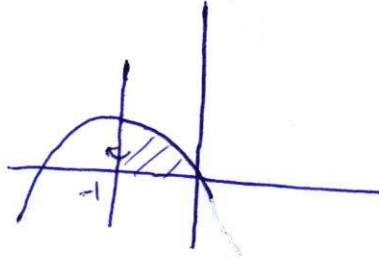
i. x-ekseni etrafında döndürülmesiyle oluşan cismin hacmini veren belirli integrali disk yöntemini kullanarak yazınız (İntegrali hesaplamayınız) (Bölgeyi çiziniz).



$$-x^2 - 2x = 0 \rightarrow x = 0 \quad x = -2$$

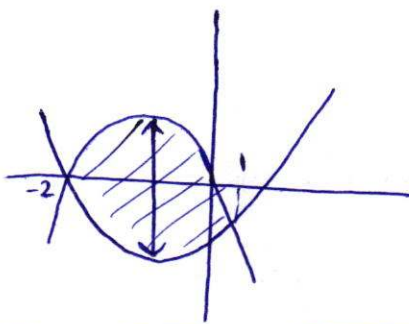
$$V = \pi \int_{-1}^0 (-x^2 - 2x)^2 dx$$

ii. $x = -1$ doğrusu etrafında döndürülmesiyle oluşan cismin hacmini veren belirli integrali silindirik kabuk yöntemini kullanarak yazınız (İntegrali hesaplamayınız) (Bölgeyi çiziniz).



$$V = 2\pi \int_{-1}^0 (x+1) \cdot (-x^2 - 2x) dx$$

b. $y = x^2 - 4$ ve $y = -x^2 - 2x$ eğrileriyle sınırlı olan D bölgesinin alanını veren belirli integrali yazınız (İntegrali hesaplamayınız) (Bölgeyi çiziniz).



$$x^2 - 4 = -x^2 - 2x \rightarrow x^2 + x - 2 = 0 \rightarrow x = 1 \quad x = -2$$

$$A = \int_{-2}^1 (-x^2 - 2x - (x^2 - 4)) dx$$

(*) $f(x) = \frac{2|x-1|}{x^2-x^3}$ fonksiyonunun süreksizlik noktalarını bulup sınıflandırın.

$$x^2 - x^3 = 0 \rightarrow x = 0 \quad x = 1$$

$$\lim_{x \rightarrow 0} \frac{2 \cdot (1-x)}{x^2(1-x)} = \infty \rightarrow \text{Sonsuz süreksiz}$$

$$\lim_{x \rightarrow 1^-} \frac{2(1-x)}{x^2(1-x)} = 2$$

$$\lim_{x \rightarrow 1^+} \frac{2(x-1)}{x^2(1-x)} = -2$$

Good Luck...

3.a. $\int_{-\infty}^{\infty} e^{-|x|} dx$ integralini hesaplayınız.

$$\int_{-\infty}^0 e^x dx + \int_0^{\infty} e^{-x} dx = \lim_{R \rightarrow -\infty} \int_R^0 e^x dx + \lim_{R \rightarrow \infty} \int_0^R e^{-x} dx$$
$$= \lim_{R \rightarrow -\infty} e^x \Big|_R^0 + \lim_{R \rightarrow \infty} -e^{-x} \Big|_0^R = \lim_{R \rightarrow -\infty} \frac{(1 - e^R)}{0} + \lim_{R \rightarrow \infty} \frac{(-e^{-R} + 1)}{0} = 2$$

~~2~~

3.b. $\int \frac{\ln(x+3) dx}{(x+1)^2}$ integralini hesaplayınız.

$$\ln(x+3) = u \quad \frac{dx}{(x+1)^2} = dv$$

$$\frac{dx}{x+3} = du \quad v = -\frac{1}{x+1}$$

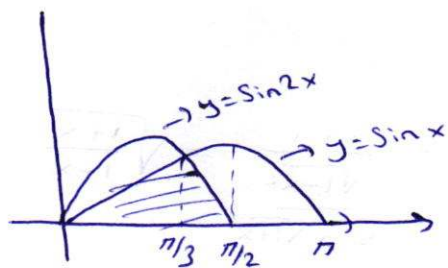
$$I = -\frac{1}{x+1} \ln(x+3) + \int \frac{dx}{(x+1)(x+3)} \Rightarrow \frac{A}{x+1} + \frac{B}{x+3} = \frac{1}{(x+1)(x+3)}$$

$$A = \frac{1}{x+3} \Big|_{x=-1} = \frac{1}{2} \quad B = \frac{1}{x+1} \Big|_{x=-3} = -\frac{1}{2}$$

$$= -\frac{1}{x+1} \ln(x+3) + \frac{1}{2} \int \left(\frac{1}{x+1} - \frac{1}{x+3} \right) dx$$

$$= -\frac{1}{x+1} \ln(x+3) + \frac{1}{2} \ln|x+1| + \frac{1}{2} \ln|x+3| + \underline{\underline{C}}$$

⑧ $0 \leq x \leq \frac{\pi}{2}$, $y = \sin x$, $y = \sin 2x$, x-ekseni arasındaki alan?



$$\sin x = \sin 2x \Rightarrow \sin x = 2 \sin x \cos x$$

$$\cos x = \frac{1}{2} \rightarrow x = \frac{\pi}{3}$$

$$A = \int_0^{\pi/3} \sin x \, dx + \int_{\pi/3}^{\pi/2} \sin 2x \, dx$$

$$= -\cos x \Big|_0^{\pi/3} + \frac{\cos 2x}{2} \Big|_{\pi/3}^{\pi/2} = \frac{3}{4}$$

⑨ $\int \frac{dx}{\sqrt{x^2+4x+8}} = \int \frac{dx}{\sqrt{(x+2)^2+4}} = \int \frac{2\sec^2\theta \, d\theta}{2\sec\theta} = \int \sec\theta \, d\theta$

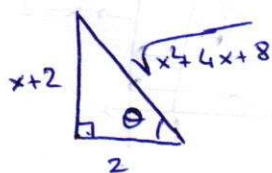
$$x+2 = 2\tan\theta$$

$$dx = 2\sec^2\theta \, d\theta$$

$$\sqrt{(x+2)^2+4} = 2\sec\theta$$

$$= \ln|\sec\theta + \tan\theta| + c$$

$$= \ln\left|\frac{\sqrt{x^2+4x+8}}{2} + \frac{x+2}{2}\right| + c$$

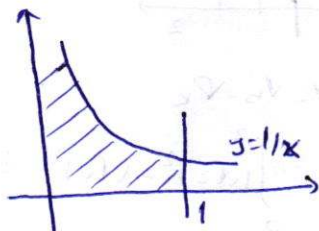


⑩ $\int_0^\infty \cos x \, dx$ integralinin karakterini belirleyiniz.

$$\int_0^\infty \cos x \, dx = \lim_{R \rightarrow \infty} \int_0^R \cos x \, dx = \lim_{R \rightarrow \infty} \sin x \Big|_0^R = \lim_{R \rightarrow \infty} \sin R \rightarrow \text{limit mevcut değildir}$$

int. İraksaktır

⑪ $y = \frac{1}{\sqrt{x}}$ in altında, x-ekseninin üstünde kalan ve $x=0$ ile $x=1$ arasında kalan bölgenin alanı?



$$A = \int_0^1 \frac{dx}{\sqrt{x}} = \lim_{c \rightarrow 0^+} \int_c^1 \frac{dx}{\sqrt{x}} = \lim_{c \rightarrow 0^+} 2\sqrt{x} \Big|_c^1$$

$$= \lim_{c \rightarrow 0^+} (2 - 2\sqrt{c}) = \underline{\underline{2}}$$

* $y = \sqrt{1-x^2}$ - ArcSin x eğrisinin $[-\frac{1}{2}, \frac{1}{2}]$ aralığındaki uzunluğunu hesaplayınız.

$$S = \int_{-1/2}^{1/2} \sqrt{1+(y')^2} dx$$

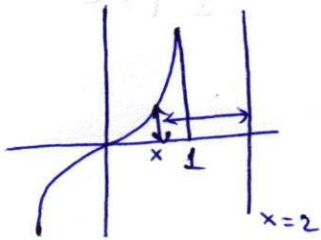
$$y' = \frac{-2x}{2\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = - \frac{x+1}{\sqrt{1-x^2}} = - \sqrt{\frac{1+x}{1-x}}$$

$$1+(y')^2 = 1 + \frac{1+x}{1-x} = \frac{2}{1-x} \Rightarrow \sqrt{1+(y')^2} = \frac{\sqrt{2}}{\sqrt{1-x}}$$

$$S = \int_{-1/2}^{1/2} \frac{\sqrt{2}}{\sqrt{1-x}} = \sqrt{2} \cdot (-2\sqrt{1-x}) \Big|_{-1/2}^{1/2} = 2(\sqrt{3}-1)$$

* $y=x^3$, $x=1$, $y=0$ arasındaki bölgenin $x=2$ etrafında çevrilmesiyle oluşan cismin hacmini veren belirli integral?

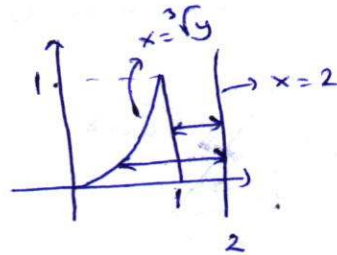
I. Yol : Kabuk yöntemi ile



$$V = 2\pi \int_0^1 (K.4or.) (K.45K.) dx$$

$$= 2\pi \int_0^1 x^3 \cdot (2-x) dx$$

II. Yol : Pul Yöntemi ile:

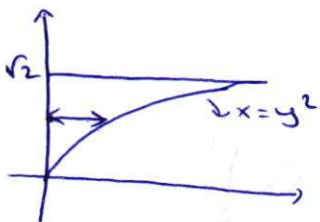


$$V = V_b - V_k$$

$$= \pi \int_0^1 (2-y^3)^2 dy - \pi \int_0^1 (2-1)^2 dy$$

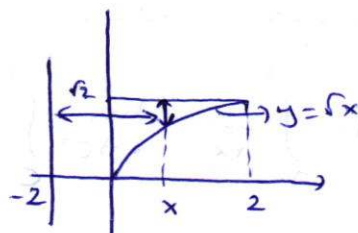
* $x=y^2$, y-ekseni, $y=\sqrt{2}$ ile sınırlı bölgenin :

a) Alanı:



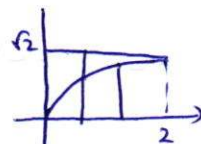
$$A = \int_0^{\sqrt{2}} y^2 dy$$

b) $x=-2$ etrafında çevrilmesi ile oluşan hacim:



$$V = 2\pi \int_0^2 (x+2) \cdot (\sqrt{2}-\sqrt{x}) dx$$

c) x-ekseni etrafında çevrilmesiyle oluşan hacim:



$$V = V_b - V_k$$

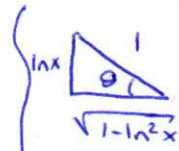
$$= \pi \int_0^2 ((\sqrt{2})^2 - (x)^2) dx$$

$$(*) \int \frac{\sqrt{1-\ln^2 x}}{x} dx = \int \cos \theta \cdot \cos \theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta$$

$$\ln x = \sin \theta$$

$$\frac{dx}{x} = \cos \theta d\theta$$

$$\sqrt{1-\ln^2 x} = \sqrt{1-\sin^2 \theta} = \cos \theta$$



$$= \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C$$

$$= \frac{1}{2} \text{Arcsin}(\ln x) + 2(\ln x)(\sqrt{1-\ln^2 x}) + C$$

(*) $\int (\ln x)^n dx$ integrali için bir indirgeme formülü bulup

bu formül yardımıyla $\int \ln^3 x dx$ integralini hesaplayınız.

$$I_n = \int (\ln x)^n dx \text{ olsun.}$$

$$(\ln x)^n = u \rightarrow \frac{n(\ln x)^{n-1}}{x} dx = du$$

$$dx = dv \rightarrow v = x$$

$$I_n = x(\ln x)^n - n \int (\ln x)^{n-1} dx \Rightarrow \boxed{I_n = x(\ln x)^n - n I_{n-1}}$$

$$\int \ln^3 x dx = I_3 = x(\ln x)^3 - 3 \cdot I_2 = x(\ln x)^3 - 3 \cdot [x(\ln x)^2 - 2 \cdot I_1]$$

$$= x(\ln x)^3 - 3x(\ln x)^2 + 6(x \ln x - x) + C$$

$$= x(\ln x)^3 - 3x(\ln x)^2 + 6x \ln x - 6x + C$$

$$\int \ln x dx = x \ln x - x$$

$$(*) \int \frac{\cos x}{\sqrt{1+\sin^2 x}} dx = \int \frac{du}{\sqrt{1+u^2}} = \int \frac{\sec^2 a}{\sec a} da = \int \sec a da = \ln |\sec a + \tan a| + C$$

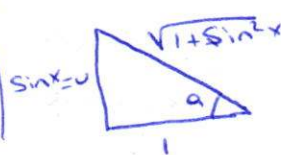
$$u = \sin x$$

$$du = \cos x dx$$

$$u = \tan a$$

$$du = \sec^2 a da$$

$$\sqrt{1+u^2} = \sec a$$



$$= \ln |\sqrt{1+\sin^2 x} + \sin x| + C$$

$$(*) \int \frac{\ln x}{x\sqrt{4+\ln^2 x}} dx = \int \frac{u}{\sqrt{4+u^2}} du = \int \frac{2 \tan t \cdot 2 \sec^2 t}{2 \sec t} dt = 2 \int \sec t \tan t dt$$

$$= 2 \sec t + C$$

$$= 2 \cdot \frac{\sqrt{4+\ln^2 x}}{2} + C$$

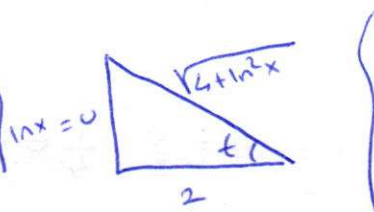
$$= \underline{\underline{\sqrt{4+\ln^2 x} + C}}$$

$$\ln x = u \quad \frac{dx}{x} = du$$

$$u = 2 \tan t$$

$$du = 2 \sec^2 t$$

$$\sqrt{4+u^2} = 2 \sec t$$



4.a. $\int_1^2 \frac{dx}{x(\ln x)^2}$ integralini hesaplayınız.

$$= \lim_{c \rightarrow 1^+} \int_c^2 \frac{dx}{x(\ln x)^2} \quad \begin{array}{l} \ln x = u \\ \frac{dx}{x} = du \end{array} \quad \left| \quad = \lim_{c \rightarrow 1^+} \int_{\ln c}^{\ln 2} \frac{du}{u^2}$$

$$= \lim_{c \rightarrow 1^+} \left[-\frac{1}{u} \right]_{\ln c}^{\ln 2} = \lim_{c \rightarrow 1^+} -\frac{1}{\ln 2} + \frac{1}{\ln c} = \infty$$

4.b. $y = \sqrt{x}$ eğrisinin $[1,2]$ aralığında kalan kısmı x-ekseni etrafında döndürülüyor. Meydana gelen yüzeyin alanını bulunuz.

$$S = 2\pi \int_1^2 y \sqrt{1+(y')^2} dx = 2\pi \int_1^2 \sqrt{x} \left(\sqrt{1+\left(\frac{1}{2\sqrt{x}}\right)^2} \right) dx$$

$$S = 2\pi \int_1^2 \sqrt{x} \cdot \frac{\sqrt{4x+1}}{2\sqrt{x}} dx = \pi \int_1^2 \sqrt{4x+1} dx$$

$$4x+1 = t^2$$

$$4dx = 2t dt$$

$$x=1 \rightarrow t=\sqrt{5}$$

$$x=2 \rightarrow t=3$$

$$S = \pi \int_{\sqrt{5}}^3 t \cdot 2t dt = \pi \left. \frac{2}{3} t^3 \right|_{\sqrt{5}}^3$$

$$S = \frac{2\pi}{3} (27 - 5\sqrt{5})$$

3.a. $\int \frac{\sec^2 x dx}{\tan x[(\sec^2 x)-2]}$ integralini hesaplayınız.

$$\tan x = u$$

$$\sec^2 x dx = du$$

$$\int \frac{du}{u(1+u^2-2)} = \int \frac{du}{u(u^2-1)}$$

$$\frac{1}{u(u-1)(u+1)} = \frac{A}{u} + \frac{B}{u-1} + \frac{C}{u+1} \quad \begin{array}{l} A=-1 \\ B=\frac{1}{2} \\ C=\frac{1}{2} \end{array}$$

$$\begin{aligned} \int \left[-\frac{1}{u} + \frac{1}{2(u-1)} + \frac{1}{2(u+1)} \right] du &= -\ln|u| + \frac{1}{2} \ln|u-1| + \frac{1}{2} \ln|u+1| + C \\ &= \ln \left| \frac{\sqrt{u^2-1}}{u} \right| + C = \ln \left| \frac{\sqrt{\tan^2 x - 1}}{\tan x} \right| + C \end{aligned}$$

3.b. $f(x) = \frac{\sqrt{1-x^2}}{x^2+4}$ ile tanımlı olan f fonksiyonuna $[-1,1]$ aralığında Rolle teoremi uygulanabilir mi?

Eğer uygulanabilirse, teoremi sağlayan c değerlerini bulunuz.

a) $f(x)$ in tanım kümesi $[-1,1]$ olduğundan bu aralıkta sürekli.

$$b) f'(x) = \frac{\frac{-2x}{2\sqrt{1-x^2}}(x^2+4) - \sqrt{1-x^2} \cdot 2x}{(x^2+4)^2} = \frac{x^3-6x}{\sqrt{1-x^2}(x^2+4)^2}$$

f $(-1,1)$ de türemlenebilir.

c) $f(-1) = 0 = f(1)$ olduğundan Rolle teoremi uygulanabilir.

$f'(c) = 0$ o.s. en az bir $c \in (-1,1)$ sayısı vardır.

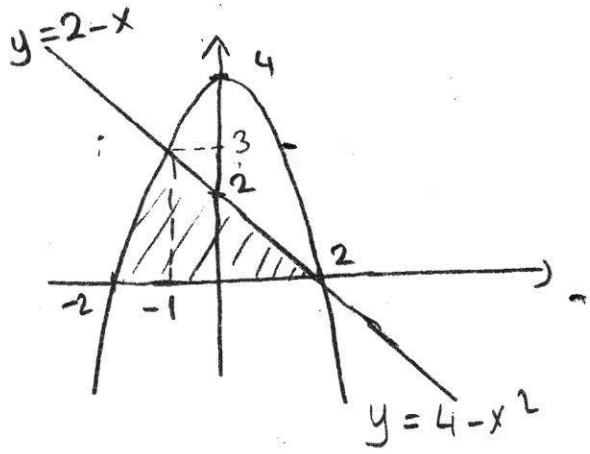
$$\frac{c^3-6c}{\sqrt{1-c^2}(c^2+4)^2} = 0 \Rightarrow c^3-6c = 0 \Rightarrow c(c^2-6) = 0$$

$$c_1 = 0 \in (-1,1)$$

$$c_2 = \sqrt{6} \notin (-1,1)$$

$$c_3 = -\sqrt{6} \notin (-1,1)$$

Soru 4-a) $y=4-x^2$ eğrisi, $x+y=2$ doğrusu ve x -ekseni ile sınırlanmış bölgenin alanını bulunuz. (12 P)



$$A = \int_{-2}^{-1} (4-x^2) dx + \int_{-1}^2 (2-x) dx$$

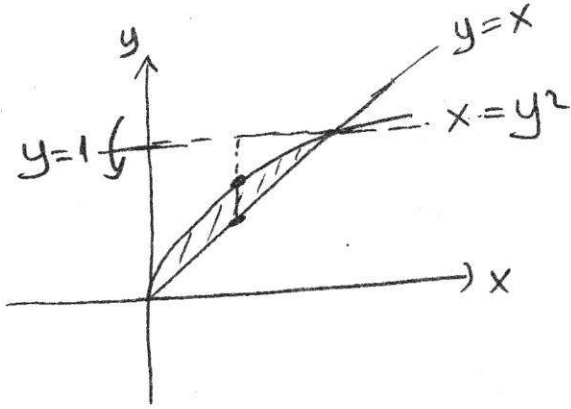
$$A = \left(4x - \frac{x^3}{3}\right) \Big|_{-2}^{-1} + \left(2x - \frac{x^2}{2}\right) \Big|_{-1}^2$$

$$A = \frac{5}{3} + \frac{9}{2} = \frac{37}{6} //$$

2°40°

$$A = \int_0^3 [(2-y) - \sqrt{4-y}] dy$$

Soru 4-b) $x=y^2$ parabolü ve $y=x$ doğrusu ile sınırlı bölgenin, $y=1$ doğrusu etrafında döndürülmesiyle oluşan cismin hacmini bulunuz. (13 P)



$$V = \pi \int_0^1 [(1-x)^2 - (1-\sqrt{x})^2] dx$$

$$V = \pi \int_0^1 (x^2 + 2\sqrt{x} - 3x) dx$$

$$V = \pi \left(\frac{x^3}{3} + \frac{4}{3} x^{\frac{3}{2}} - \frac{3}{2} x^2 \right) \Big|_0^1 = \frac{\pi}{6} //$$

2°40°

$$A = 2\pi \int_0^1 \left(\begin{matrix} \text{Kobuk} \\ \text{Yarık} \end{matrix} \right) \left(\begin{matrix} \text{Kobuk} \\ \text{Yükseklik} \end{matrix} \right) dy$$

$$A = 2\pi \int_0^1 (1-y)(y-y^2) dy = 2\pi \int_0^1 (y^3 - 2y^2 + y) dy = \frac{\pi}{6} //$$

S 3.a) $y = \ln(\cos x)$ eğrisinin $0 \leq x \leq \frac{\pi}{3}$ aralığındaki yay uzunluğunu hesaplayınız. (15p)

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = - \frac{\sin x}{\cos x} = -\tan x \quad (2)$$

$$L = \int_0^{\pi/3} \sqrt{1 + \tan^2 x} dx = \int_0^{\pi/3} |\sec x| dx = \int_0^{\pi/3} \sec x dx \quad (4) \quad (2)$$

$$L = \ln |\sec x + \tan x| \Big|_0^{\pi/3} = \ln |\sec \frac{\pi}{3} + \tan \frac{\pi}{3}| - \ln |\sec 0 + \tan 0| \quad (5)$$
$$= \ln (2 + \sqrt{3}) - \ln 1 \quad (2)$$
$$= \ln (2 + \sqrt{3}) \text{ br.}$$

b) $f(x) = \begin{cases} \frac{1 - \cos 2x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$
(10p)

ile tanımlı f fonksiyonunun $x = 0$ daki sürekliliğini araştırınız.

$$\lim_{x \rightarrow 0} f(x) = f(0) = 1 \text{ olmalı.}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} = 2 \neq 1 = f(0) \quad (3) \quad (2)$$

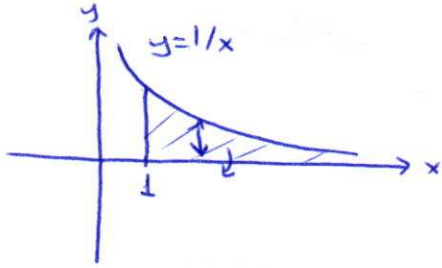
olduğundan f , $x=0$ da sürekli değildir. (2)

$$(*) f(x) = \begin{cases} 1/\sqrt{x} & , 0 < x \leq 1 \\ x-1 & , 1 < x \leq 2 \end{cases}$$

$$\Rightarrow \int_0^2 f(x) dx = ?$$

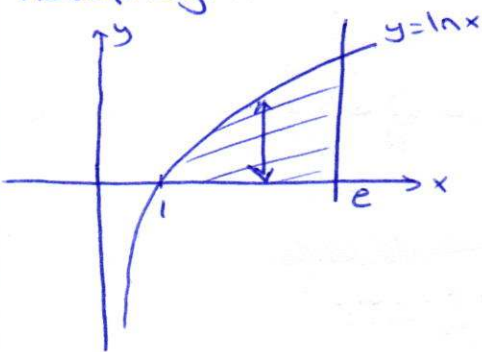
$$\begin{aligned} \int_0^2 f(x) dx &= \int_0^1 \frac{dx}{\sqrt{x}} + \int_1^2 (x-1) dx = \lim_{c \rightarrow 0^+} \int_c^1 \frac{dx}{\sqrt{x}} + \left(\frac{x^2}{2} - x \right) \Big|_1^2 \\ &= \lim_{c \rightarrow 0^+} (2\sqrt{x}) \Big|_c^1 + (2-2-\frac{1}{2}+1) \\ &= \lim_{c \rightarrow 0^+} (2-2\sqrt{c}) + \frac{1}{2} = \underline{\underline{\frac{5}{2}}} \end{aligned}$$

⊛ $x=1$ doğrusunun sağında, kalan ve $y=\frac{1}{x}$ ve $y=0$ ile sınırlı bölgenin x -ekseni etrafında döndürülmesiyle oluşan hacim?



$$\begin{aligned} V &= \pi \int_1^{\infty} \left(\frac{1}{x} \right)^2 dx = \pi \cdot \lim_{R \rightarrow \infty} \int_1^R \frac{dx}{x^2} \\ &= \pi \cdot \lim_{R \rightarrow \infty} \left(-\frac{1}{x} \right) \Big|_1^R = \pi \cdot \lim_{R \rightarrow \infty} \left(-\frac{1}{R} + 1 \right) = \underline{\underline{\pi}} \end{aligned}$$

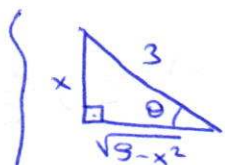
⊛ $y=\ln x$, $x=e$, x -ekseni ile sınırlı bölgenin x -ekseni etrafında döndürülmesiyle oluşan hacmi disk yöntemiyle hesaplayın.



$$\begin{aligned} V &= \pi \int_1^e (\ln x)^2 dx & u &= (\ln x)^2 \rightarrow du = 2 \ln x \cdot \frac{1}{x} dx \\ &= \pi \left[x(\ln x)^2 \Big|_1^e - 2 \int_1^e \ln x dx \right] & dx &= dv \rightarrow v = x \\ &= \pi \left[e - 2 \left[x \ln x \Big|_1^e - \int_1^e dx \right] \right] & \ln x &= u \rightarrow \frac{dx}{x} = du \\ &= \pi \left[e - 2e + 2x \Big|_1^e \right] = \underline{\underline{\pi(e-2)}} & dx &= dv \rightarrow v = x \end{aligned}$$

$$\textcircled{*} \int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{3 \cos \theta} 3 \cos \theta d\theta = 9 \int \sin^2 \theta d\theta = \frac{9}{2} \int (1 - \cos 2\theta) d\theta$$

$$\begin{aligned} x &= 3 \sin \theta \\ dx &= 3 \cos \theta d\theta \\ \sqrt{9-x^2} &= 3 \cos \theta \end{aligned}$$



$$= \frac{9}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) + C$$

$$= \frac{9}{2} \left(\text{ArcSin} \frac{x}{3} - 2 \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} \right) + C$$

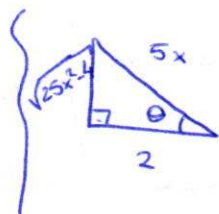
$$\textcircled{*} \int \frac{dx}{\sqrt{25x^2-4}} = \frac{1}{5} \int \frac{2 \sec \theta \tan \theta}{2 \tan \theta} d\theta = \frac{1}{5} \int \sec \theta d\theta = \frac{1}{5} \ln |\sec \theta + \tan \theta| + C$$

$$5x = 2 \sec \theta$$

$$5dx = 2 \sec \theta \tan \theta d\theta$$

$$\sqrt{25x^2-4} = \sqrt{4(\sec^2 \theta - 1)}$$

$$= 2 \tan \theta$$



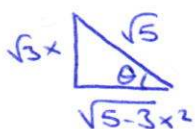
$$= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2} \right| + C$$

$$\textcircled{*} \int \sqrt{5-3x^2} dx = \int \sqrt{5} \cos \theta \cdot \frac{\sqrt{5}}{3} \cos \theta d\theta = \frac{5}{3} \int \frac{1 + \cos 2\theta}{2} d\theta$$

$$\sqrt{3}x = \sqrt{5} \sin \theta$$

$$\sqrt{3}dx = \sqrt{5} \cos \theta d\theta$$

$$\sqrt{5-3x^2} = \sqrt{5} \cos \theta$$



$$= \frac{5}{6} \theta + \frac{5}{6} \cdot \frac{\sin 2\theta}{2} + C$$

$$= \frac{5}{6} \text{ArcSin} \frac{\sqrt{3}x}{\sqrt{5}} + \frac{10}{12} \cdot \frac{\sqrt{3}x}{\sqrt{5}} \cdot \frac{\sqrt{5-3x^2}}{\sqrt{5}} + C$$

$$\textcircled{*} \int e^{-x} \ln(1+e^x) dx = -e^{-x} \ln(1+e^x) + \int \frac{dx}{1+\frac{e^x}{e^{-x}}}$$

$$\ln(1+e^x) = u$$

$$e^{-x} dx = du$$

$$\downarrow$$

$$\frac{e^x}{1+e^x} dx = du$$

$$\downarrow$$

$$-e^{-x} = v$$

$$= -e^{-x} \ln(1+e^x) + \int \frac{e^{-x}}{1+e^x} dx$$

$$1+e^{-x} = u$$

$$-e^{-x} dx = du$$

$$= -e^{-x} \ln(1+e^x) - \ln|1+e^{-x}| + C$$

S 3.a) $\int_1^{e^{\pi/4}} \frac{dx}{x \cos^2(\ln x)}$ integralini hesaplayınız.

(12 p)

$$\left. \begin{aligned} \ln x &= u \\ \frac{1}{x} dx &= du \end{aligned} \right\} \textcircled{3} \quad \left. \begin{aligned} x &= 1 \text{ için } u = \ln 1 = 0 \\ x &= e^{\pi/4} \text{ için } u = \ln e^{\pi/4} = \frac{\pi}{4} \end{aligned} \right\} \textcircled{2}$$

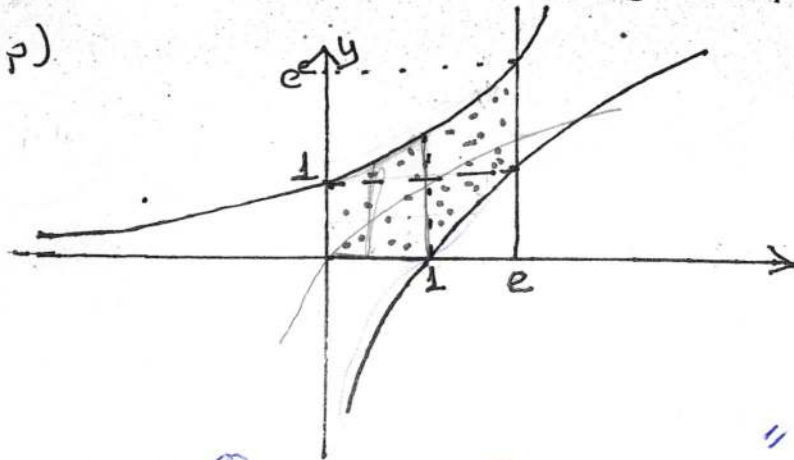
$$\int_1^{e^{\pi/4}} \frac{dx}{x \cos^2(\ln x)} = \int_0^{\pi/4} \frac{du}{\cos^2 u} = \int_0^{\pi/4} \sec^2 u \, du \quad \textcircled{3}$$

$$= \tan u \Big|_0^{\pi/4} \textcircled{3} = \tan \frac{\pi}{4} - \tan 0$$

$$= 1 \quad \textcircled{1}$$

b) $y = \ln x$, $y = e^x$ eğrilerinin $x = 0$, $y = 0$ ve $x = e$ doğrularıyla sınırladığı bölgenin (şeklini çizerek) alanını x 'e bağlı belirli integral ile ifade ediniz. **İntegrali hesaplamayınız.**

(13 p)



⑬

$$= \int_0^e e^x \, dx - \int_0^e \ln x \, dx$$

$$i) A = \int_0^1 e^x \, dx + \int_1^e (e^x - \ln x) \, dx$$

⑤

$$ii) A = \int_0^1 e^y \, dy + \int_1^e (e - \ln y) \, dy$$

⑤

$$b) \int_0^e e^y \, dy - \int_0^e \ln y \, dy$$

(*) $3y^2 = x^3$ eğrisinin $A(1, \frac{1}{\sqrt{3}})$, $B(3, 3)$ noktaları

arasındaki uzunluğunu bulunuz.

$$S = \int_1^3 \sqrt{1+(y')^2} dx$$

$$3y^2 = x^3$$

↓ Türev

$$3 \cdot 2y \cdot y' = 3x^2$$

$$y' = \frac{x^2}{2y}$$

$$(y')^2 = \frac{x^4}{4y^2} \xrightarrow{4y^2 \rightarrow \frac{x^3}{3}} = \frac{x^4}{4 \cdot \frac{x^3}{3}} = \frac{3}{4} x$$

$$1 + (y')^2 = 1 + \frac{3x}{4} = \frac{4+3x}{4}$$

$$\sqrt{1+(y')^2} = \frac{\sqrt{4+3x}}{2}$$

$$S = \int_1^3 \frac{\sqrt{4+3x}}{2} dx = \int_7^{13} \frac{\sqrt{u}}{6} du = \frac{1}{6} \cdot \frac{u^{3/2}}{\frac{3}{2}} \Big|_7^{13}$$

$$= \frac{1}{9} [(13)^{3/2} - (7)^{3/2}]$$

$$4+3x = u$$

$$3dx = du$$

$$x=1 \rightarrow u=7$$

$$x=3 \rightarrow u=13$$

(*) $\lim_{x \rightarrow 0} \frac{\int_0^x \frac{\sin t}{\cos t} dt}{\tan x} = ? \frac{0}{0} \Rightarrow \text{L'Hopital}$

$$\lim_{x \rightarrow 0} \frac{\frac{\sin^2 x}{\cos x}}{\sec^2 x} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\cos^3 x} \cdot \cos^2 x = \boxed{0}$$

* $9y^2 = 4(1+x^2)^3$ eğrisinin $x=0$ $x=3$ arasındaki uzunluğu?

$$9y^2 = 4(1+x^2)^3 \rightarrow 9 \cdot 2yy' = 12(1+x^2)^2 \cdot 2x$$

$$y' = \frac{4x(1+x^2)^2}{3y}$$

$$1 + (y')^2 = 1 + \frac{16x^2(1+x^2)^4}{9y^2} = 1 + 4x^2(1+x^2) = 4x^4 + 4x^2 + 1 = (2x^2 + 1)^2$$

$$\sqrt{1 + (y')^2} = \sqrt{(2x^2 + 1)^2} = 2x^2 + 1$$

$$S = \int_0^3 \sqrt{1 + (y')^2} dx = \int_0^3 (2x^2 + 1) dx = \underline{\underline{21}}$$

* $x = 2\sqrt{4-y}$ eğrisinin $0 \leq y \leq \frac{15}{4}$ arasındaki kısmının y-ekseni etrafında serpilmesiyle oluşan cismin yüzey alanını veren integrali yazınız.

$$V = 2\pi \int_0^{15/4} x \cdot \sqrt{1 + (x')^2} dy$$

$$x = 2\sqrt{4-y} \rightarrow (x') = \frac{-1}{\sqrt{4-y}} \rightarrow \sqrt{1 + (x')^2} = \sqrt{1 + \frac{1}{4-y}} = \sqrt{\frac{5-y}{4-y}}$$

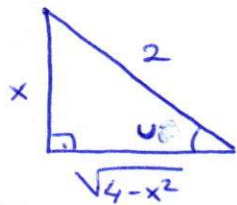
$$S = 2\pi \int_0^{15/4} 2 \cdot \sqrt{4-y} \cdot \frac{\sqrt{5-y}}{\sqrt{4-y}} dy = 4\pi \int_0^{15/4} \sqrt{5-y} dy$$

$$(*) \int \frac{\sqrt{4-x^2}}{x} dx = \int \frac{2 \cos u}{2 \sin u} \cdot 2 \cos u du = 2 \int \frac{\cos^2 u}{\sin u} du = 2 \int \frac{1 - \sin^2 u}{\sin u} du$$

$$x = 2 \sin u$$

$$dx = 2 \cos u du$$

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2 u} = 2 \cos u$$



$$= 2 \int \operatorname{cosec} u du - 2 \int \sin u du$$

$$= -2 \ln |\operatorname{cosec} u + \cot u| + 2 \cos u$$

$$= -2 \ln \left| \frac{\sqrt{4-x^2}}{x} + \frac{2}{x} \right| + \frac{2\sqrt{4-x^2}}{2} + C$$

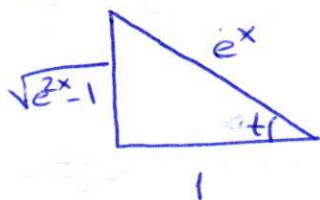
$$(*) \int \frac{dx}{e^{2x} - e^x} = \int \frac{du}{u(u^2 - u)} = \int \frac{du}{u^2(u^2 - 1)} = \int \frac{\operatorname{Sect} t \cdot \operatorname{Tant}}{\operatorname{Sec}^2 t \cdot (\operatorname{Sec}^2 t - 1)} dt$$

$$e^x = u$$

$$e^x dx = du \rightarrow dx = \frac{du}{u}$$

$$\left\{ \begin{array}{l} u = \operatorname{Sect} t \\ du = \operatorname{Sect} t \operatorname{Tant} dt \end{array} \right\} = \int \frac{dt}{\operatorname{Sect} t \cdot \operatorname{Tant}} = \int \frac{\cos^2 t}{\sin t} dt$$

$$u = e^x = \operatorname{Sect} t$$



$$= \int \operatorname{cosec} t dt - \int \sin t dt$$

$$= -\ln |\operatorname{cosec} t + \cot t| + \cos t + C$$

$$= -\ln \left| \frac{e^x}{\sqrt{e^{2x}-1}} + \frac{1}{\sqrt{e^{2x}-1}} \right| + \frac{1}{e^x} + C$$

$$(*) \int \sin x \cdot \ln(\tan x) dx = -\cos x \cdot \ln(\tan x) + \int \cos x \cdot \frac{1}{\sin x \cos x} dx$$

$$\left\{ \begin{array}{l} \ln(\tan x) = u \\ \sin x dx = dv \\ \frac{\sec^2 x}{\tan x} dx = du \\ v = -\cos x \end{array} \right.$$

$$= -\cos x \ln(\tan x) + \int \operatorname{cosec} x dx$$

$$= -\cos x \ln(\tan x) - \ln |\cot x| + \operatorname{cosec} x + C$$

$$\downarrow$$

$$\frac{1}{\sin x \cos x} dx = du$$

$$(*) \int \frac{dx}{x^3 + 9x} = \int \frac{1}{9x} dx - \frac{1}{9} \int \frac{x}{x^2 + 9} dx = \frac{1}{9} \ln|x| - \frac{1}{9} \cdot \frac{1}{2} \ln|x^2 + 9| + C$$

$$\frac{1}{x(x^2 + 9)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 9} = \frac{Ax^2 + 9A + Bx^2 + Cx}{x(x^2 + 9)}$$

$$\left\{ \begin{array}{l} A + B = 0 \\ C = 0 \\ 9A = 1 \end{array} \right. \quad \left\{ \begin{array}{l} A = \frac{1}{9} \\ B = -\frac{1}{9} \end{array} \right.$$

④ $\int_0^1 x \ln x \, dx$ integralini $\cos\pi$ sonucu yorumlayınız.

$$\int_0^1 x \ln x \, dx = \lim_{c \rightarrow 0^+} \int_c^1 x \ln x \, dx = \lim_{c \rightarrow 0^+} \left[\frac{x^2}{2} \ln x \Big|_c^1 - \int_c^1 \frac{x}{2} \, dx \right]$$

$$\begin{aligned} \ln x = u &\rightarrow \frac{dx}{x} = du \\ x \, dx = dv &\rightarrow v = \frac{x^2}{2} \end{aligned} \quad \left\{ \begin{aligned} &= \lim_{c \rightarrow 0^+} \left[-\frac{c^2}{2} \ln c - \frac{x^2}{4} \Big|_c^1 \right] \\ &= \lim_{c \rightarrow 0^+} \left[-\frac{\ln c}{\frac{2}{c^2}} \right] - \lim_{c \rightarrow 0^+} \left(\frac{1}{4} - \frac{c^2}{4} \right) \end{aligned} \right.$$

$$= \lim_{c \rightarrow 0^+} \left(-\frac{\frac{1}{c}}{\frac{-4}{c^3}} \right) - \frac{1}{4} = \lim_{c \rightarrow 0^+} \frac{1}{4} c^2 - \frac{1}{4} = -\frac{1}{4}$$

$$\int_0^1 x \ln x \, dx = -\frac{1}{4} \Rightarrow \text{integral yakınsaktır.}$$

⑤ $\int_0^\infty \frac{\arctan x}{1+x^2} \, dx = ?$

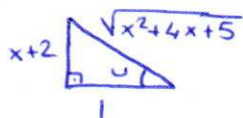
$$\int_0^\infty \frac{\arctan x}{1+x^2} \, dx = \lim_{R \rightarrow \infty} \int_0^R \frac{\arctan x}{1+x^2} \, dx = \lim_{R \rightarrow \infty} \int_0^{\arctan R} u \, du$$

$$\begin{aligned} \arctan x &= u \\ \frac{dx}{1+x^2} &= du \\ x=R &\Rightarrow u = \arctan R \\ x=0 &\Rightarrow u=0 \end{aligned} \quad \left\{ \begin{aligned} &= \lim_{R \rightarrow \infty} \frac{u^2}{2} \Big|_0^{\arctan R} \\ &= \lim_{R \rightarrow \infty} \frac{(\arctan R)^2}{2} = \frac{1}{2} \cdot \left(\frac{\pi}{2} \right)^2 \\ &= \frac{\pi^2}{8} \end{aligned} \right.$$

$$\textcircled{*} \int \frac{dx}{\sqrt{x^2+4x+5}} = \int \frac{dx}{\sqrt{(x+2)^2+1}} = \int \frac{\sec^2 u \, du}{\sqrt{\tan^2 u + 1}} = \int \frac{\sec^2 u}{\sec u} du$$

$$x^2+4x+5 = x^2+4x+4+1 \quad \left\{ \begin{array}{l} x+2 = \tan u \\ dx = \sec^2 u \, du \end{array} \right. \quad = \int \sec u \, du$$

$$= (x+2)^2+1$$



$$= \ln |\sec u + \tan u| + C$$

$$= \ln |\sqrt{x^2+4x+5} + x+2| + C$$

$$\textcircled{*} \int x^3 e^{x^2} dx = \frac{1}{2} \int t \cdot e^t dt = \frac{t e^t}{2} - \frac{1}{2} \int e^t dt = \frac{t e^t}{2} - \frac{e^t}{2} + C$$

$$x^2 = t \quad 2x dx = dt \quad \left\{ \begin{array}{l} t = u \rightarrow dt = du \\ e^x dt = dv \rightarrow v = e^t \end{array} \right. \quad = \frac{x^2 e^{x^2}}{2} - \frac{e^{x^2}}{2} + C$$

$$\textcircled{*} \int \frac{x^{1/2}}{4(1+x^{3/4})} dx = \int \frac{u^2}{4(1+u^3)} \cdot 4u^3 du = \int \frac{u^3+1-1}{1+u^3} \cdot u^2 du$$

$$x = u^4$$

$$dx = 4u^3 du$$

$$= \int u^2 du - \int \frac{u^2}{1+u^3} du$$

$$= \frac{u^3}{3} - \frac{1}{3} \ln |1+u^3| + C$$

$$= \frac{x^{3/4}}{3} - \frac{1}{3} \ln |1+x^{3/4}| + C$$

$$\textcircled{*} \int \frac{6x+7}{(x+2)^2} dx = \int \left(\frac{6}{x+2} - \frac{5}{(x+2)^2} \right) dx = 6 \ln |x+2| + \frac{5}{x+2} + C$$

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$$

$$= \frac{Ax+2A+B}{(x+2)^2}$$

$$\left. \begin{array}{l} A = 6 \\ 2A+B = 7 \end{array} \right\} \begin{array}{l} A = 6 \\ B = -5 \end{array}$$

$$(*) I = \int \frac{\text{ArcSin } 2x}{x^2 \sqrt{1-4x^2}} dx = ?$$

I. Adım: Değişken dönüşümü

$$2x = u \quad 2dx = du$$

$$I = \frac{1}{2} \int \frac{\text{ArcSin } u}{\frac{u^2}{4} \sqrt{1-u^2}} du = 2 \int \frac{\text{ArcSin } u}{u^2 \sqrt{1-u^2}}$$

$$= 2 \int \frac{\overbrace{\text{ArcSin}(\sin \theta)}^{\theta}}{\sin^2 \theta \cdot \cancel{\cos \theta}} \cdot \cancel{\cos \theta} d\theta$$

$$= 2 \int \theta \cdot \text{Cosec}^2 \theta d\theta$$

$$= 2 \left[-\theta \cot \theta + \int \cot \theta d\theta \right]$$

$$= 2 \left[-\theta \cot \theta + \ln |\sin \theta| \right] + c$$

$$= -2 \cdot \text{ArcSin } 2x \cdot \frac{\sqrt{1-4x^2}}{2x} + 2 \ln |\sin 2x| + c$$

2. Adım: Trig. dönüşüm

$$u = \sin \theta$$

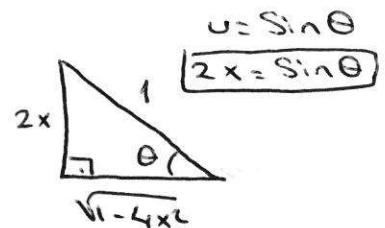
$$du = \cos \theta d\theta$$

$$\sqrt{1-u^2} = \cos \theta$$

3. Adım: Kısmi int.

$$u = \theta \quad dv = \text{Cosec}^2 \theta d\theta$$

$$du = d\theta \quad v = -\cot \theta$$



Nasıl Soru Ama;))

Korkmayın sınavda bu kadar zor sormayız:))